Memo (8/7)

Note Title 7/5/2020

1 (a) Show that $\langle 6,9,-3\rangle$ and $\langle -4,-6,2\rangle$ are parallel.

$\langle 6,9,-3\rangle = 3\langle 2,3,-1\rangle$

$\langle -4,-6,2\rangle = -2\langle 2,3,-1\rangle$

$\therefore$ Both vectors are parallel to $\langle 2,3,-1\rangle$

$\Rightarrow \langle 6,9,-3\rangle \parallel \langle -4,-6,2\rangle$.

OR $\frac{1}{3}\langle 6,9,-3\rangle = -\frac{1}{2}\langle -4,-6,2\rangle$

$\therefore 2\langle 6,9,-3\rangle = -3\langle -4,-6,2\rangle$

$\therefore \langle 6,9,-3\rangle = -3/2\,\langle -4,-6,2\rangle$

$\Rightarrow \langle 6,9,-3\rangle \parallel \langle -4,-6,2\rangle$.

ALSO: $\langle 6,9,-3\rangle \times \langle -4,-6,2\rangle$

$$= \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ 6 & 9 & -3 \\ -4 & -6 & 2 \end{vmatrix} = \bar{i}(18-18) - \bar{j}(12-12) + \bar{k}(-36+36)$$

$$= \langle 0,0,0\rangle$$

$\Rightarrow |\langle 6,9,-3\rangle \times \langle -4,-6,2\rangle|$

$= 0 = |\langle 6,9,-3\rangle|\,|\langle -4,-6,2\rangle| \sin\theta$

$\Rightarrow \sin\theta = 0 \Rightarrow \theta = 0$ or $\theta = \pi$

$\Rightarrow$ 2 vectors are parallel.

(1b) Find a vector $\bar{a}$ that is orthogonal to $\langle -4,-6,2\rangle$.
Show that $\bar{a}$ is orthogonal to $\langle -4,-6,2\rangle$.

Suppose $\langle a,b,c\rangle \perp \langle -4,-6,2\rangle = 0$

Many possibilities
Must test!

$-4a-6b+2c = 0$

$\Rightarrow 2a+3b-c = 0$

$\Rightarrow$ If $a=3, b=-2, c=0$: $\langle 3,-2,0\rangle \cdot \langle -4,-6,0\rangle$
$= -12+12+0$
$= 0$ NB.

$\Rightarrow \langle 3,-2,0\rangle \perp \langle -4,-6,2\rangle$. $\Rightarrow \bar{a} = \langle 3,-2,0\rangle$.

(1c) Find a unit vector $\bar{a}$ in the direction opposite to $\langle -4,-6,2\rangle$.

$$\bar{a} = -\frac{\langle -4,-6,2\rangle}{|\langle -4,-6,2\rangle|}$$

$|\langle -4,-6,2\rangle| = \sqrt{16+36+4} = \sqrt{56}$

$$= \frac{\langle 4,6,-2\rangle}{\sqrt{56}}$$

$$= \frac{1}{\sqrt{56}}\langle 4,6,-2\rangle = \left\langle \frac{4}{\sqrt{56}}, \frac{6}{\sqrt{56}}, \frac{-2}{\sqrt{56}} \right\rangle$$

Q2: Find the value(s) of x such that the angle between $\langle 0,1,-1\rangle$ and $\langle -1,x,0\rangle$ is $2\pi/3$.

$$\langle 0,1,-1\rangle \cdot \langle -1,x,0\rangle = |\langle 0,1,-1\rangle|\,|\langle -1,x,0\rangle| \cos\frac{2\pi}{3}$$

$$\langle 0,1,-1\rangle \cdot \langle -1,x,0\rangle = 0 + x + 0 = x$$

$$\Rightarrow x = \sqrt{2}\sqrt{1+x^2}\cos 2\pi/3$$
$$= \sqrt{2}\sqrt{1+x^2}\left(-\tfrac{1}{2}\right)$$

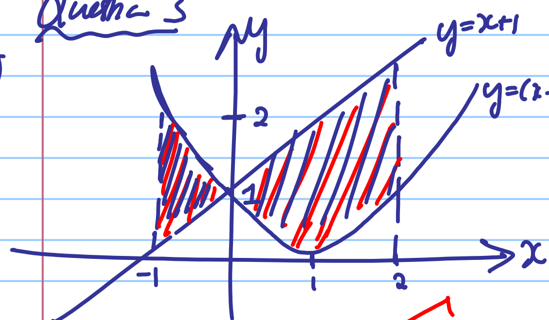
$(x<0!)$

$$\Rightarrow x^2 = 2(1+x^2)\times\tfrac{1}{4}$$
$$\Rightarrow 2x^2 = 1+x^2$$
$$\Rightarrow x^2 = 1$$
$$\Rightarrow x^2-1 = 0 \Rightarrow \underline{x=-1} \quad (\text{since } x<0).$$

-1 if they have $x=\pm 1$.

Question 3

(a)

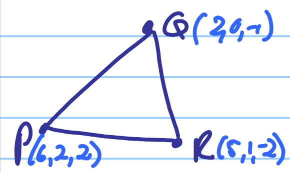


$-\frac{1}{2}$ for each mistake

(b) $$A = \int_{-1}^{0} (x-1)^2-(x+1)\,dx + \int_{0}^{2} (x+1)-(x-1)^2\,dx$$

Question 4:

(a)



$$\overline{PQ} = \langle 2,0,-1\rangle - \langle 6,2,2\rangle = \langle -4,-2,-3\rangle$$
$$\overline{PR} = \langle 5,1,-2\rangle - \langle 6,2,2\rangle = \langle -1,-1,-4\rangle$$
$$\overline{QR} = \langle 5,1,-2\rangle - \langle 2,0,-1\rangle = \langle 3,1,-1\rangle$$

Directions NB.

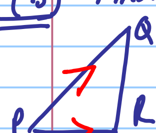
If they have $\vec{u},\vec{v},\vec{w}$, they must show directions.

$$\overline{PQ}\cdot\overline{PR} = \overline{PR}\cdot\overline{PQ}$$
$$= \langle -1,-1,-4\rangle\cdot\langle -4,-2,-3\rangle$$
$$= 4+8+12 = 24.$$

$$\overline{RP}\cdot\overline{RQ} = \overline{RQ}\cdot\overline{RP}$$
$$= \langle -3,-1,1\rangle\cdot\langle 1,1,4\rangle$$
$$= -3-1+4$$
$$= 0$$

$$\Rightarrow \overline{RP}\perp\overline{RQ} \Rightarrow \hat{R} = \frac{\pi}{2} \Rightarrow \triangle PRQ \text{ is a right triangle.}$$

(b) Find area of triangle PQR



$$\text{Area } \triangle PQR = \frac{1}{2}\left|\overline{PQ}\times\overline{PR}\right|$$

$$\overline{PQ} \times \overline{PR} = \langle -4,-2,-3\rangle \times \langle -1,-1,-4\rangle$$

$$= \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ -4 & -2 & -3 \\ -1 & -1 & -4 \end{vmatrix}$$

$$= \bar{i}(8-3) - \bar{j}(16-3) + \bar{k}(4-2)$$

$$= 5\bar{i} - 13\bar{j} + 2\bar{k}$$

$$= \langle 5,-13,2\rangle$$

$$\begin{array}{r} 169 \\ 29 \\ \hline 198 \end{array}$$

Area triangle is $\frac{1}{2}|\langle 5,-13,2\rangle|$

$$= \frac{1}{2}\sqrt{25+169+4}$$

$$= \frac{1}{2}\sqrt{198}$$

<u>or</u> Area triangle $= \frac{1}{2}|\overline{RP} \times \overline{RQ}| = \frac{1}{2}|\overline{RQ} \times \overline{RP}|$

<u>or</u> Area triangle $= \frac{1}{2}|\overline{QR} \times \overline{QP}| = \frac{1}{2}|\overline{QP} \times \overline{QR}|$.

<u>Also correct</u>: Δ triangle $= \frac{1}{2} \times$ base $\times \perp$ height.